Amabwiriza y'umubaji: imbere muri sitidiyo

Ibyo umubaji akeneye byose ngo atunganye imbere muri kontineri ya 20 ft ibe icyumba cy'abashyitsi (sitidiyo), intambwe ku yindi, n'ibipimo byose.

> ***Ubusobanuro:*** *iyi nyandiko yahinduwe na mudasobwa (machine draft). Igomba gusuzumwa n'umuntu uvuga Ikinyarwanda kavukire mbere yo kuyiha abakozi. Iyo hari itandukaniro, kurikiza iy'Icyongereza.*

Status: verisiyo A2, 2026-10-28, PRELIMINARY. Ibipimo biva ku mafoto kugeza igihe hapimwe ku Munsi wa 1; atelier ntigira icyo ikata mbere ya verisiyo B.

Sitidiyo zombi zirasa, bityo aya mabwiriza akora kuri zombi.

English: read this guide in English · Gucapa: aya mabwiriza nka PDF

Amagambo akoreshwa hano

- **End A (impera A)**: urukuta rufunze rw'impera, iruhande rw'igitanda. **End B (impera B)**: impera ifite inzugi za kontineri, iruhande rw'ubwiherero.
- **Back wall (urukuta rw'inyuma)**: urukuta rurerure rudafite aho gufungukira (igitanda na sofa). **Front wall (urukuta rw'imbere)**: urukuta rurerure rufite urugi runyerera rw'ikirahure n'idirishya rito W1.
- **From the steel (uhereye ku cyuma)**: bipimwe uhereye ku cyuma cyambaye ubusa, mbere yo gushyiraho lining. **From the finished wall (uhereye ku rukuta rurangiye)**: bipimwe uhereye ku isura ya lining.
- Ibipimo byose ni milimetero (mm) keretse handitswe ukundi.

Sitidiyo mu ncamake

Iki	Ingano (mm)
Imbere muri kontineri, icyuma ku cyuma	uburebure 5898, ubugari 2352, hejuru 2393
Icyumba kirangiye, lining kuri lining	uburebure 5798, ubugari 2252, hejuru 2316 (13.1 m²)
Uko urukuta rwubakwa	50: igiti (batten) 38 gifite spray foam hagati, hanyuma pulayiwudi (plywood) 12
Uko igisenge cyubakwa	60

Iki	Ingano (mm)
Uko hasi hubakwa	17: impapuro za pulayiwudi 12, hanyuma imbaho z'igiti (oak planks) 5 (mu bwiherero: vinyl)
Urugi runyerera rw'ikirahure, uhereye ku rukuta rurangiye rwa end A	kuva 1380 kugeza 4380, hejuru kuri 2240
Partition y'ubwiherero, uhereye ku rukuta rurangiye rwa end A	itangirira kuri 4578, ubugari 70

Ninde	Intambwe
Umubaji, abantu 2	S02, S03, S05 (lining), S06, S07, S11 kugeza S18
Umusuderi (welder)	S01 (gusudira kose)
Umuyagankuba (electrician) n'umufundi w'amazi (plumber)	S04 (first fix), S19 (second fix)
Ushyiraho spray foam	S05 (foam)
Ushyira amakaro (tiler)	S08

Ingingo eshanu zo guhagarara ukagenzura (hold points)

1. **Ibipimo byagarutse.** Ibipimo byo ku Munsi wa 1 byageze, verisiyo B irasohoka. Kugeza icyo gihe atelier ntigira icyo ikata; imirimo yo ku kibanza S01 kugeza S03 irakomeza.
2. **Gusudira byarangiye.** Gusudira no gusya (grinding) kose (S01) birangira mbere y'uko foam ishyirwamo.
3. **Igerageza ry'umuvuduko ryatsinze.** First fix (S04) igeragezwa kuri bar 10, ibisubizo bikandikwa mbere ya foam.
4. **Foam yumye.** Foam imaze amasaha 24 yuma kandi yaciwe ingana n'ibiti mbere y'urupapuro rwa mbere rwa lining.
5. **Inkuta z'amazi zagenzuwe.** Membrane y'ubwiherero igenzurwa mbere y'uko ushyira amakaro atangira.

Amategeko abiri adahinduka:

- Ntuzigere ukuba (sand), uharura (plane) cyangwa ukata hasi ha kontineri: hatewe umuti wica udukoko. Hapfuke na membrane, hanyuma ushyireho impapuro nshya ukoresheje vis.
- Pima uhereye ku cyuma nk'uko imbonerahamwe zibivuga, hanyuma ugenzure ibipimo birangiye ukurikije aya mabwiriza.

Umunsi wa 1: pima kontineri

Pima kontineri zombi za sitidiyo mbere y'ikindi cyose, wohereze imibare uwo munsi. Andika **U1** na **U2** ku nzugi zazo ukoresheje ingwa (chalk). Kontineri ya gatatu ni ibiro (office); ntipimwa hano.

Urupapuro rw'ibipimo, uzuza inkingi ya value_mm · source `study/studio/survey.csv`

unit	item	value_mm	measured_by	date	note
U1	int_length				inside length, end wall to end wall, at 1 m height
U1	int_width				inside width at 1 m height, mid-length
U1	int_height				floor to roof at mid-width
U1	o1_start				slider opening start from the inside face of end A
U1	o1_end				slider opening end from the inside face of end A
U1	o2_start				narrow door opening start from end A
U1	o2_end				narrow door opening end from end A
U1	diag_diff				floor diagonals difference
U1	level_diff				highest minus lowest corner casting
U2	int_length				inside length, end wall to end wall, at 1 m height
U2	int_width				inside width at 1 m height, mid-length
U2	int_height				floor to roof at mid-width
U2	o1_start				slider opening start from the inside face of end A
U2	o1_end				slider opening end from the inside face of end A
U2	o2_start				narrow door opening start from end A

unit	item	value_mm	measured_by	date	note
U2	o2_end				narrow door opening end from end A
U2	diag_diff				floor diagonals difference
U2	level_diff				highest minus lowest corner casting

1. **int_length**: uburebure bw'imbere, kuva ku cyuma cya end A kugeza ku cyuma cya end B, kuri m 1 hejuru ya hasi.
2. **int_width**: ubugari bw'imbere, icyuma ku cyuma, kuri m 1 hejuru ya hasi, hagati mu burebure.
3. **int_height**: kuva hasi kugeza ku cyuma cy'igisenge, hagati mu bugari.
4. **o1_start** na **o1_end**: aho umwanya munini w'urugi runyerera utangirira n'aho urangirira, uhereye ku cyuma cya end A.
5. **o2_start** na **o2_end**: aho urugi rwa kera rufunganye rutangirira n'aho rurangirira, uhereye ku cyuma cya end A.
6. **diag_diff**: pima diyagonale zombi za hasi, inguni ku nguni; andika itandukaniro.
7. **level_diff**: ukoresheje level y'amazi, inguni (corner casting) iri hejuru ukuyemo iri hasi.
8. Fata amafoto: buri rukuta ururebe imbere, hasi, inguni enye (corner castings), ikadiri y'urugi runyerera, urugi rwa kera rufunganye.

Uko imirimo ikurikirana

Intambwe	Umurimo	Ninde	Hold point
S01	Gupima, kuringaniza aho kontineri yicaye, gusudira kose	Umusuderi n'umubaji	1, 2
S02	Membrane n'impapuro za hasi	Umubaji	
S03	Ibiti (battens)	Umubaji	
S04	First fix: insinga, conduits, amatiyo	Umuyagankuba n'umufundi w'amazi	3
S05	Spray foam, hanyuma impapuro za lining	Ushyiraho foam, hanyuma umubaji	4
S06 kugeza S10	Ubwiherero: partition, urugi, inkuta z'amazi, tray, WC, heater, lavabo	Umubaji, ushyira amakaro, umufundi w'amazi	5

Intambwe	Umurimo	Ninde	Hold point
S11 kugeza S14	Imbaho z'igiti (oak planks), hanyuma igikoni	Umubaji	
S15 kugeza S17	Igitanda, akameza ko ku buriri, intebe	Umubaji	
S18	Trims, caulk, irangi, insect screens	Umubaji n'usiga irangi	
S19	Second fix n'icyemezo (certificate)	Umuyagankuba	
S20 kugeza S27	Terase yo hejuru n'esikariye, nyuma	Umusuderi n'umubaji	

S01: gupima, kuringaniza no gusudira kose

1. Uzuza urupapuro rw'ibipimo (Umunsi wa 1 hejuru).
2. Shyira buri nguni muri enye (corner castings) ku gitereko cya beto (concrete block stand) cya 40 × 60 cm, uringanize n'ibyuma (shims); inguni iri ku rutare nayo ihabwa igitereko cyayo. Gerageza ko itandukaniro hagati y'inguni iri hejuru n'iri hasi riba munsi ya mm 5, uryandike muri level_diff.
3. Umusuderi: sudira icyuma cya mm 2 mu rugi rwa kera rufunganye (kuva 4990 kugeza 5510 uhereye ku rukuta rurangiye rwa end A, kugeza kuri 1150 hejuru), ushyireho idirishya rya louvre 520 × 890 hejuru yacyo.
4. Umusuderi: funga kandi ushyireho ingufuri ku nzugi za kontineri za end B, ukate umwobo wa mm 75 wa AC mu rukuta rw'inyuma.
5. Guhera kuri S05, nta gusudira cyangwa gusya kuri kontineri: byatwika foam.

S02: membrane n'impapuro za hasi

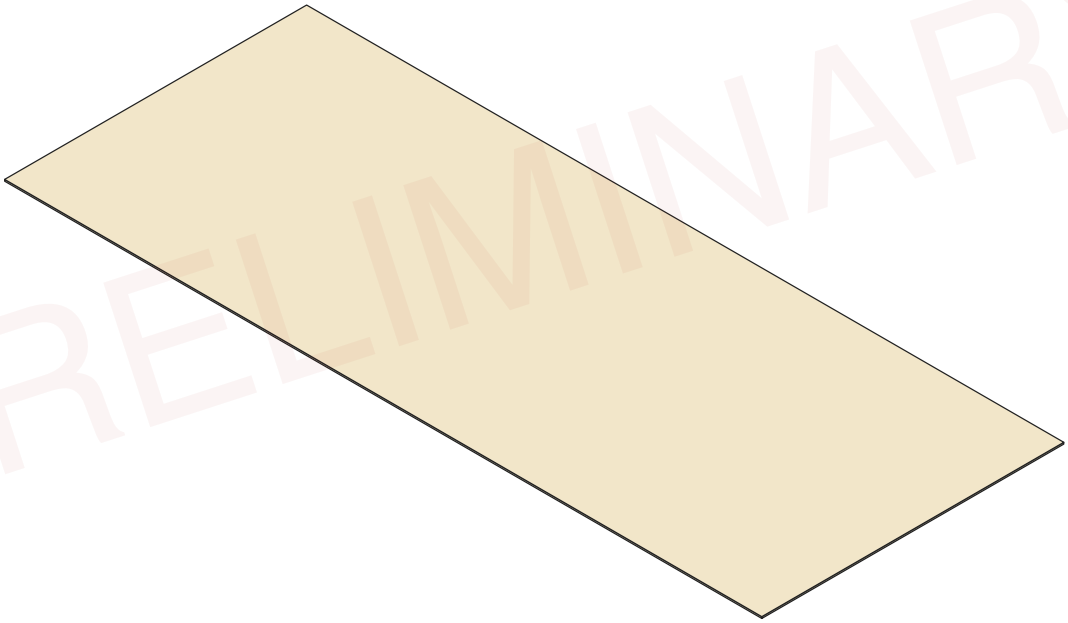
1. Siga rust converter, hanyuma zinc primer, ku cyuma cyambaye ubusa cyangwa cyafashwe n'ingese ku nkuta no ku gisenge (L 4 kuri sitidiyo).
2. Kubura hasi ha kontineri. Shyiraho isashi ya membrane (250 µm) hasi hose: ibice byegeranye bigahurira kuri mm 150 (laps), bifatanyijwe na tape, bizamurwa mm 50 ku nkuta (15.3 m² harimo laps).
3. Shyiraho impapuro za pulayiwudi 12 mm (2440 × 1220), uruhande rurerure rukurikiye kontineri, umwanya wa mm 3 hagati y'impapuro na mm 5 uvuye ku cyuma: impapuro 5 kuri sitidiyo.
4. Umurongo wa 1, ukurikiye urukuta rw'inyuma: urupapuro rwuzuye, urundi rwuzuye, hanyuma urwa nyuma rukatwa ngo rukwire (uburebure bwa hafi 1000).

5. Umurongo wa 2, ukurikiye urukuta rw'imbere, ukatwa ku bugari busigaye (hafi 1120): tangira n'igice cy'urupapuro (uburebure 1220, kivuye ku gisigazwa cy'umurongo wa 1), hanyuma urupapuro rwuzuye, hanyuma urwa nyuma rukatwa ngo rukwire (uburebure bwa hafi 2220). Nta guhurira (joint) guhuye n'ukundi ko mu murongo wa 1.
6. Bifunge ukoresheje vis z'igiti 4 × 30 (bag F) buri mm 300 mu mpande zombi, hafi 189 kuri sitidiyo. Koresha vis za mm 30 gusa, kugira ngo zigume muri hasi ha kontineri ha mm 28.
7. Genzura: nta ntera iri ku guhurira na kumwe, kandi hasi harangana kugeza kuri mm 3 munsi y'umutambiko ugororotse wa m 2.

02

Insulation and lining

Prime the steel; lay the membrane over the treated floor; screw down t



! Never sand the treated container floor. Sheets 2440 x 1220, long side along the box, joint

screw

189x F

Farm container studios farm.databayt.org/docs/studio

Prime the steel; lay the membrane over the treated floor; screw

AS-02

Scale NTS · Rev A2 · 2026-10-28 · mm

PRELIMINARY

PRELIMINARY

AS-02 Membrane and floor sheets · open full size

S03: ibiti (battens)

1. Koresha ibiti byatewe umuti 38 × 50, bihagaze ku mpapuro za hasi: 32 kuri sitidiyo.
2. Shyira igiti kimwe munsi ya buri guhurira kw'impapuro za lining, n'ikindi hagati yabyo, kugira ngo nta mwanya urenga mm 600. Shyira akamenyetso hagati ya buri giti uhereye ku cyuma, nk'uko imbonerahamwe ibivuga.
3. Fata buri giti ku cyuma ukoresheje vis zifite amababa zicukura ubwazo 5.5 × 50 (bag L): imwe kuri buri mpera n'imwe buri mm 600, 192 kuri sitidiyo.
4. Hejuru y'urugi runyerera, shyiraho igiti kimwe gitambitse ku mutwe wacyo kuri 2240 (ibice bibiri bya 1500).
5. Genzura: buri giti gihagaze neza (plumb), kandi byose biri ku rwego rumwe: umutambiko ugororotse ushyizwe ku biti bitatu ukora kuri byose uko ari bitatu.

Aho ibiti bijya, hagati ya buri giti · download CSV

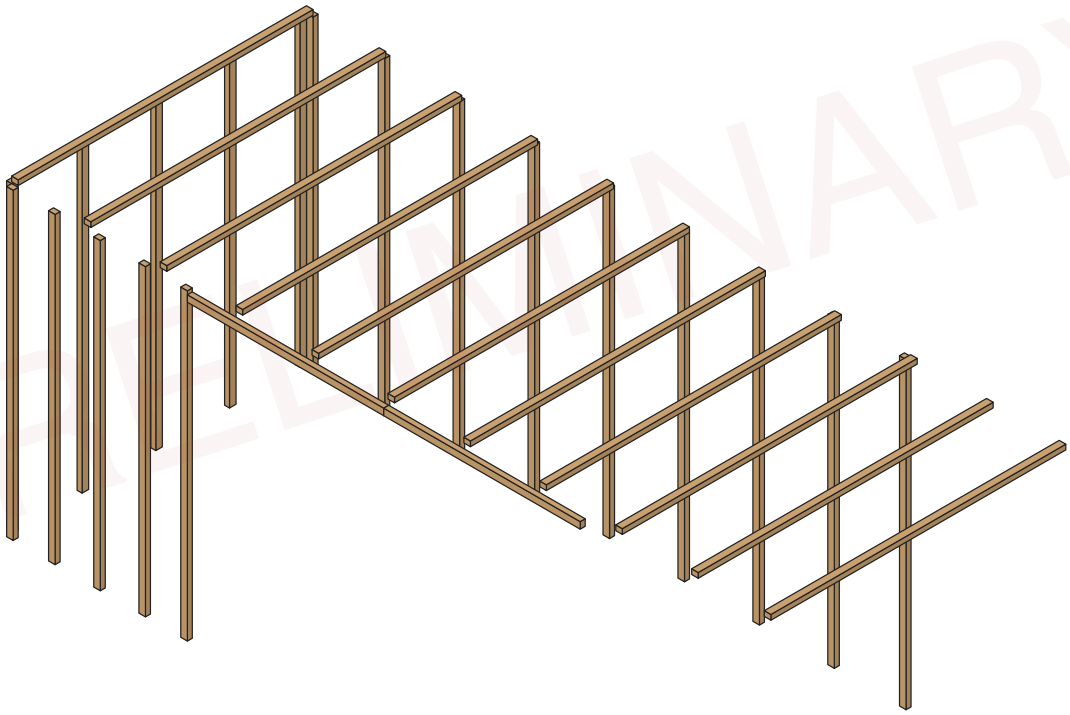
id	wall	measured	centre_mm	length_mm
K1-BAT-01	back wall	from the steel of end A	75	2316
K1-BAT-02	back wall	from the steel of end A	622	2316
K1-BAT-03	back wall	from the steel of end A	1194	2316
K1-BAT-04	back wall	from the steel of end A	1767	2316
K1-BAT-05	back wall	from the steel of end A	2339	2316
K1-BAT-06	back wall	from the steel of end A	2911	2316
K1-BAT-07	back wall	from the steel of end A	3484	2316
K1-BAT-08	back wall	from the steel of end A	4056	2316
K1-BAT-09	back wall	from the steel of end A	4603	2316
K1-BAT-10	end A	from the steel of the back wall	75	2316
K1-BAT-11	end A	from the steel of the back wall	613	2316
K1-BAT-12	end A	from the steel of the back wall	1176	2316
K1-BAT-13	end A	from the steel of the back wall	1739	2316
K1-BAT-14	end A	from the steel of the back wall	2277	2316
K1-BAT-15	front wall	from the steel of end A	75	2316
K1-BAT-16	front wall	from the steel of end A	395	2316

id	wall	measured	centre_mm	length_mm
K1-BAT-17	front wall	from the steel of end A	740	2316
K1-BAT-18	front wall	from the steel of end A	1085	2316
K1-BAT-19	front wall	from the steel of end A	1405	2316
K1-BAT-20	slider head	horizontal, from the steel of end A	1430-2930	1500
K1-BAT-21	slider head	horizontal, from the steel of end A	2930-4430	1500
K1-BAT-22	ceiling	from the steel of end A	75	2252
K1-BAT-23	ceiling	from the steel of end A	630	2252
K1-BAT-24	ceiling	from the steel of end A	1210	2252
K1-BAT-25	ceiling	from the steel of end A	1789	2252
K1-BAT-26	ceiling	from the steel of end A	2369	2252
K1-BAT-27	ceiling	from the steel of end A	2949	2252
K1-BAT-28	ceiling	from the steel of end A	3529	2252
K1-BAT-29	ceiling	from the steel of end A	4109	2252
K1-BAT-30	ceiling	from the steel of end A	4688	2252
K1-BAT-31	ceiling	from the steel of end A	5268	2252
K1-BAT-32	ceiling	from the steel of end A	5823	2252

03

Insulation and lining

Fix the battens: one under every lining-panel joint and edge, infill a



! Winged screws into the steel at both ends and every 600 mm along each batten.

sds

192x L

Farm container studios farm.databayt.org/docs/studio

Fix the battens: one under every lining-panel joint and edge, in **AS-03**

Scale NTS · Rev A2 · 2026-10-28 · mm **PRELIMINARY**

PRELIMINARY

S04: first fix, ikorwa n'umuyagankuba n'umufundi w'amazi

1. Gumana imyanya iri hagati y'ibiti isukuye kandi yumye.
2. Bafashe gushyira ibimenyetso ku dusanduku (boxes) n'inzira z'amatiyo ukurikije ibishushanyo E-101 na P-101 mbere y'uko insinga zishyirwamo.
3. Ntutangire S05 kugeza igerageza ry'umuvuduko kuri bar 10 ryanditswe kandi rigasinywaho.

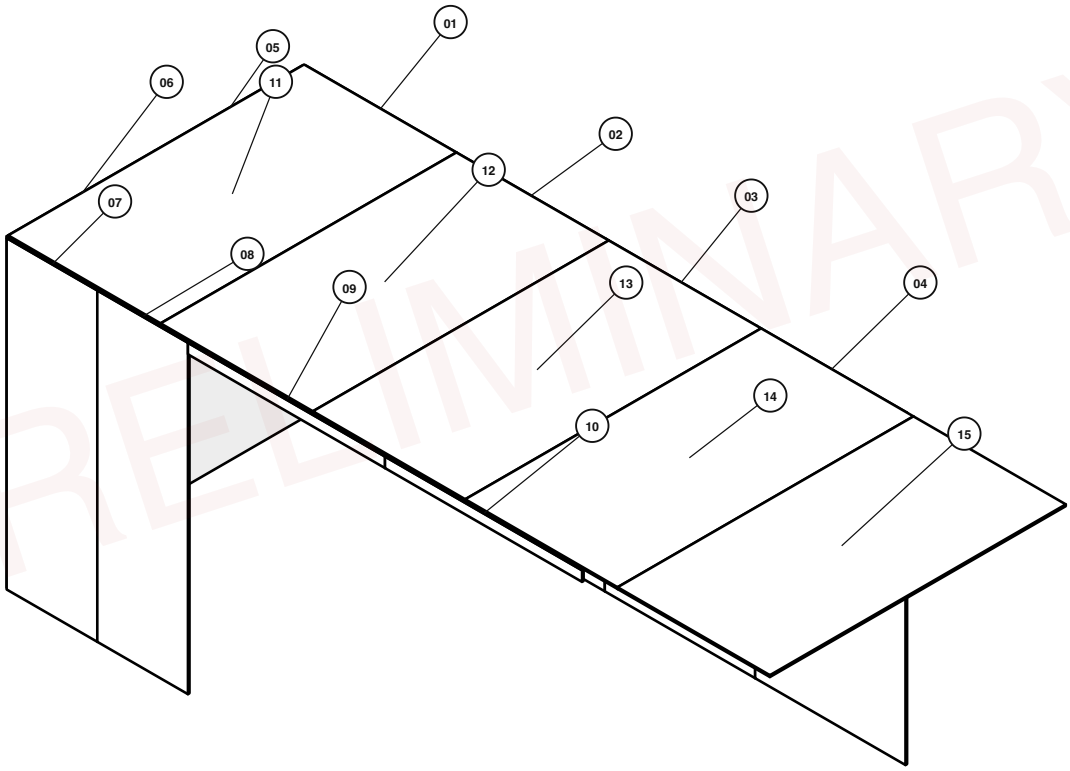
S05: spray foam, hanyuma impapuro za lining

1. Ushyiraho spray foam yuzuza hagati y'ibiti foam ifunze (closed-cell), hafi mm 38, ku nkuta no ku gisenge: 34.9 m² kuri sitidiyo. Banza upfuke ikadiri y'urugi runyerera, idirishya na hasi.
2. Nyuma y'amasaha 24, kata foam ingane n'ibiti.
3. Shyiraho impapuro 15 za lining zifite nimero (pulayiwudi 12, nimero K1-01 kugeza K1-15), banza igisenge, hanyuma inkuta. Buri ruhande rw'urupapuro rugwa hagati ku giti.
4. Fata ukoresheje vis z'igiti 4 × 30 buri mm 200 ku mpande na buri mm 300 hagati, hafi 600 kuri sitidiyo.
5. Genzura buri mwobo watunganyijwe (cut-out) ugereranyije n'agasanduku kawo mbere yo gufunga urupapuro; hanyuma ugenzure ko nta foam igaragara kandi isura z'impapuro ziri ku rwego rumwe.

05

Insulation and lining

Spray foam between the battens, let it cure and trim it flush; fix the



! HOLD: the S04 pressure test has passed. No welding or grinding on the shell once the foam

screw

600x F

Farm container studios	farm.databayt.org/docs/studio
Spray foam between the battens, let it cure and trim it flush; fix the	**AS-05**
Scale NTS · Rev A2 · 2026-10-28 · mm	**PRELIMINARY**

PRELIMINARY

AS-05 Spray foam and lining panels · open full size

S06 kugeza S10: ubwiherero (kit K2)

1. S06: shinga partition (ubugari 70, kuri 4578 uhereye ku rukuta rurangiye rwa end A) n'urukuta rwambukiranya (cross wall), ufunge isura zabyo.
2. S07: shyiraho ikadiri y'urugi umanike urugi rw'ubwiherero (ubugari 600, rufungukira mu bwiherero).
3. S08, ushyira amakaro: cement board ya mm 12 na membrane itinjiza amazi ku nkuta z'amazi, hanyuma amakaro 300 × 600 uhereye hasi ujya hejuru; vinyl y'umutekano y'ibara ry'ivu, urupapuro rumwe, hasi mu bwiherero, izamuka mm 100 ku nkuta (coved).
4. S09: shyiraho tray yo kwiyuhagiriramo (1000 × 900), ledge yayo n'ikirahure gihamye (screen).
5. S10: manika WC ifatwa ku rukuta, agasanduku ka water heater ka L 30 (450 × 450 × 300) ku rukuta rwa end B hejuru ya WC, lavabo n'akabati k'indorerwamo mu gice cyo gukarabiramo.

S11 kugeza S14: imbaho z'igiti, hanyuma igikoni (kit K3)

1. S11: shyiraho imbaho z'igiti (oak planks) (11.1 m² harimo 8% y'inyongera) hasi mu gice cyo kubamo, uhereye kuri end A kugeza kuri partition: imirongo ikurikiye kontineri, impera z'imbaho zitandukanye nibura mm 300 n'iz'umurongo ubanza.
2. S11, hanyuma: shinga fridge bay (550), hob base (600) na sink base (450) ku maguru yazo ukurikije partition: umurongo umwe wa 1600, ubujyakuzimu 600.
3. S12: teranya akabati ko hejuru ukamanike ku cyuma (steel rail), abantu babiri.
4. S13: tunganya kandi ufate worktop 38 (ubujyakuzimu 620), shyiramo sink na hob, fata plinth.
5. S14: fata hood shelf munsi y'akabati ko hejuru umanike hood.
6. Genzura: inzira kuva kuri worktop kugeza ku mpera ya sofa ni nibura 650 (yateganyijwe 658).

S15 kugeza S18: igitanda, akameza ko ku buriri, intebe, trims

1. S15, igitanda (kit K4): fata headboard ku rukuta rw'inyuma, ufunge ikadiri na bolts, ushyireho imbaho 20 (slats) ukoresheje vis, winjize udusanduku tubiri two kubikamo. Igitanda kiva kuri end A kugeza kuri 1640.
2. S16, akameza ko ku buriri (kit K5): fatanya amaguru na kole, ufate hejuru n'urubaho rwo hasi na vis; gahagarara kuri 1640 kugeza 1980.
3. S17, intebe (kit K6): shyira sofa (2000 kugeza 3300) uteranye ameza n'intebe ebyiri.
4. S18, trims (kit K7): tunganya kandi ufate trims, shyiramo caulk, siga irangi ry'umweru (off-white satin), shyiraho insect screens.

Atelier: gukata ibice bya kits

Ni nyuma ya verisiyo B gusa, ivuye ku bipimo.

1. Gura impapuro z'ibiti za sitidiyo zombi icyarimwe (batch imwe). Pima impapuro icumi: pulayiwudi 18 igomba kuba 18.0 ± 0.3 mm, bitabaye ibyo bimenyeshe uwakoze igishushanyo mbere yo gukata.
2. Kata mu mirongo (± 0.5 mm), hanyuma ukate mu bice; andika nimero ya buri gice ku isura itagaragara ako kanya. Impapuro nini: diyagonale zingana kugeza kuri mm 1.
3. Shyiraho edge band aho cut list ibigaragaza: L1 ni uruhande rumwe rurerure, L2S2 ni impande zose.
4. Cukura ukoresheje templates: banza ucape T-01 kuri 100% ugenzure akabari kayo ka mm 100.
5. Teranya by'igerageza igikoni n'igitanda bya sitidiyo 1 muri atelier mbere yo gupakira.
6. Pakira ipaki imwe kuri buri kit kuri buri sitidiyo, itarenze hafi kg 25, yanditseho "Studio n / Kit / bundle x of y".

Kuri sitidiyo: ibice 133, biri muri **cut list**, hamwe n'ibishushanyo byo gukata C-101 kugeza C-107 na dosiye za CNC.

Ibyo kugura, kuri sitidiyo

Ikintu	Umubare
Pulayiwudi 12, impapuro 2440 × 1220, hasi	5
Pulayiwudi 12, impapuro 2440 × 1220, lining	15
Pulayiwudi 18, impapuro 2440 × 1220	8
Pulayiwudi itinjiza amazi (marine) 18, impapuro 2440 × 1220 (ubwiherero)	3
Worktop 38, 3050 × 650	1
Ibiti byatewe umuti 38 × 50, uburebure m 2.4	32, wongereho 10%
Isashi ya membrane 250 µm	15.3 m²
Imbaho z'igiti (oak planks) 5 zifite underlay	11.1 m²
Spray foam, ishyirwaho n'umuhanga wayo	34.9 m²
Vis zifite amababa 5.5 × 50 (bag L)	192
Vis z'igiti 4 × 30 (bag F), hasi na lining	189 + 600
PU foam, amacupa ya ml 750	12

Ikintu	Umubare
ABS edge band	m 161

Urutonde rwuzuye n'ibiciro ni bill of materials.

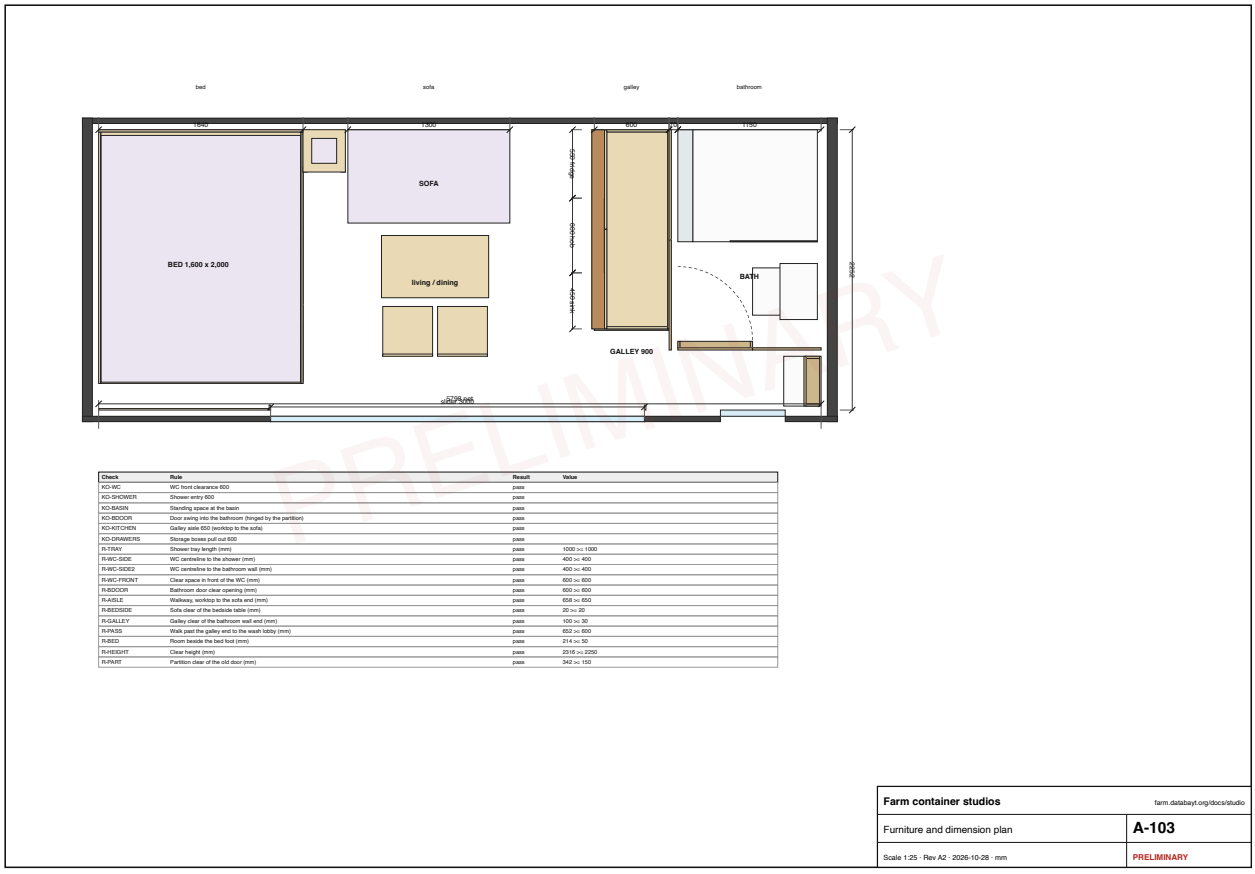
Mbere yo kuvuga ko byarangiye

1. Inzugi n'utubati dukururwa bigenda neza, urugi rw'ubwiherero rufunguka nta kirubangamira, urugi runyerera rufungwa, insect mesh irareze.
2. Nta foam igaragara, buri guhurira kw'impapuro kuri ku giti, nta mutwe wa vis usohoka.
3. Imyanya isigara: inzira ku mpera ya sofa nibura 650, 600 imbere ya WC, 600 aho binjirira muri douche.
4. Igishushanyo ubwacyo cyatsinze igenzura ryacyo 33 (checks).

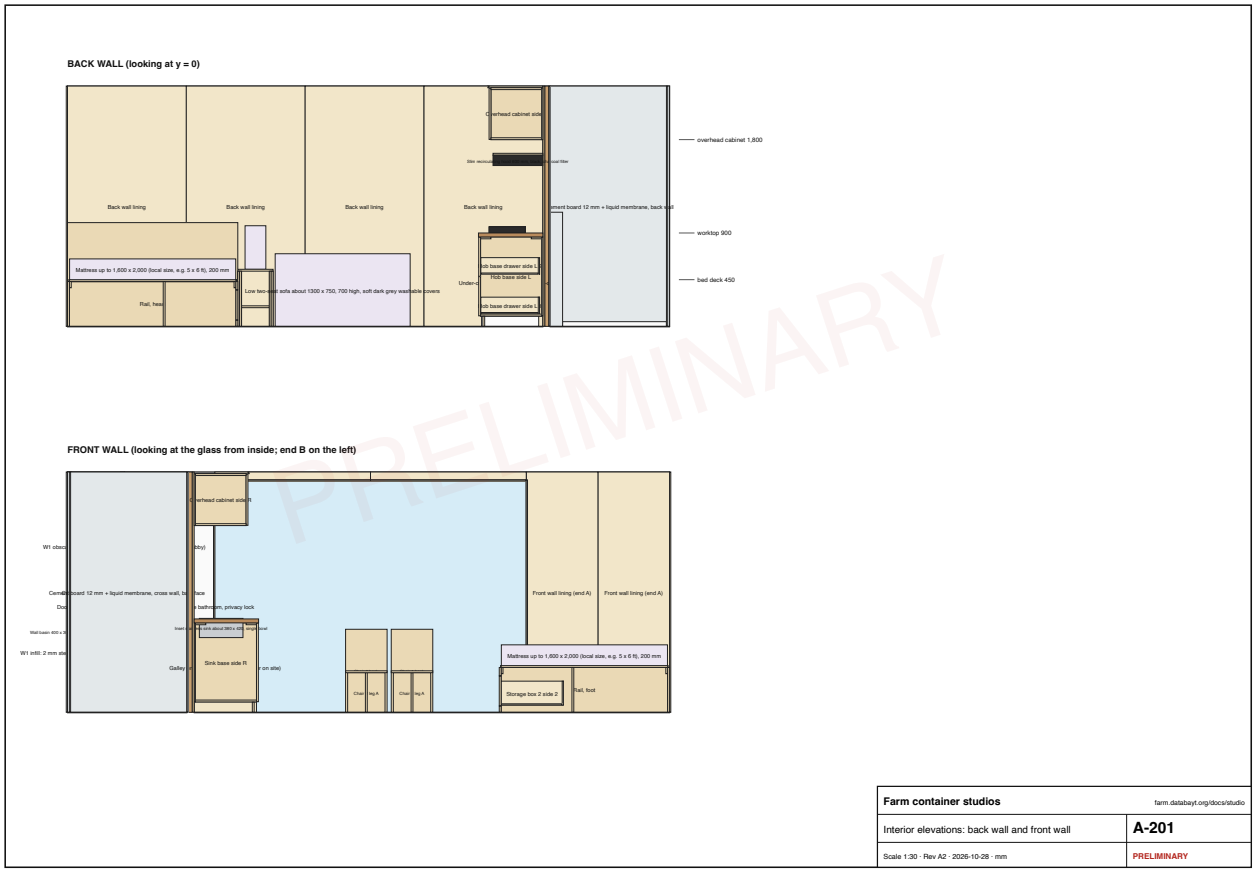
Ibibazo bigomba gusubizwa mbere y'intambwe ivugwa

1. Ni izihe kontineri ebyiri ziba U1 na U2? Mbere y'uko Umunsi wa 1 urangira.
2. Hari umusuderi wateganyijwe kuri S01? Mbere y'uko Umunsi wa 1 urangira.
3. Ni nde uzashyiraho spray foam? Nta muhanga wa spray foam i Kigali uremezwa. Mbere ya S05.
4. Ubunini bw'impapuro za hasi: mm 12, keretse ibipimo bigaragaje ko hasi ha kera hatarangana birenze mm 5 kuri m 2; icyo gihe ni mm 18 na verisiyo nshya. Mbere ya S02.

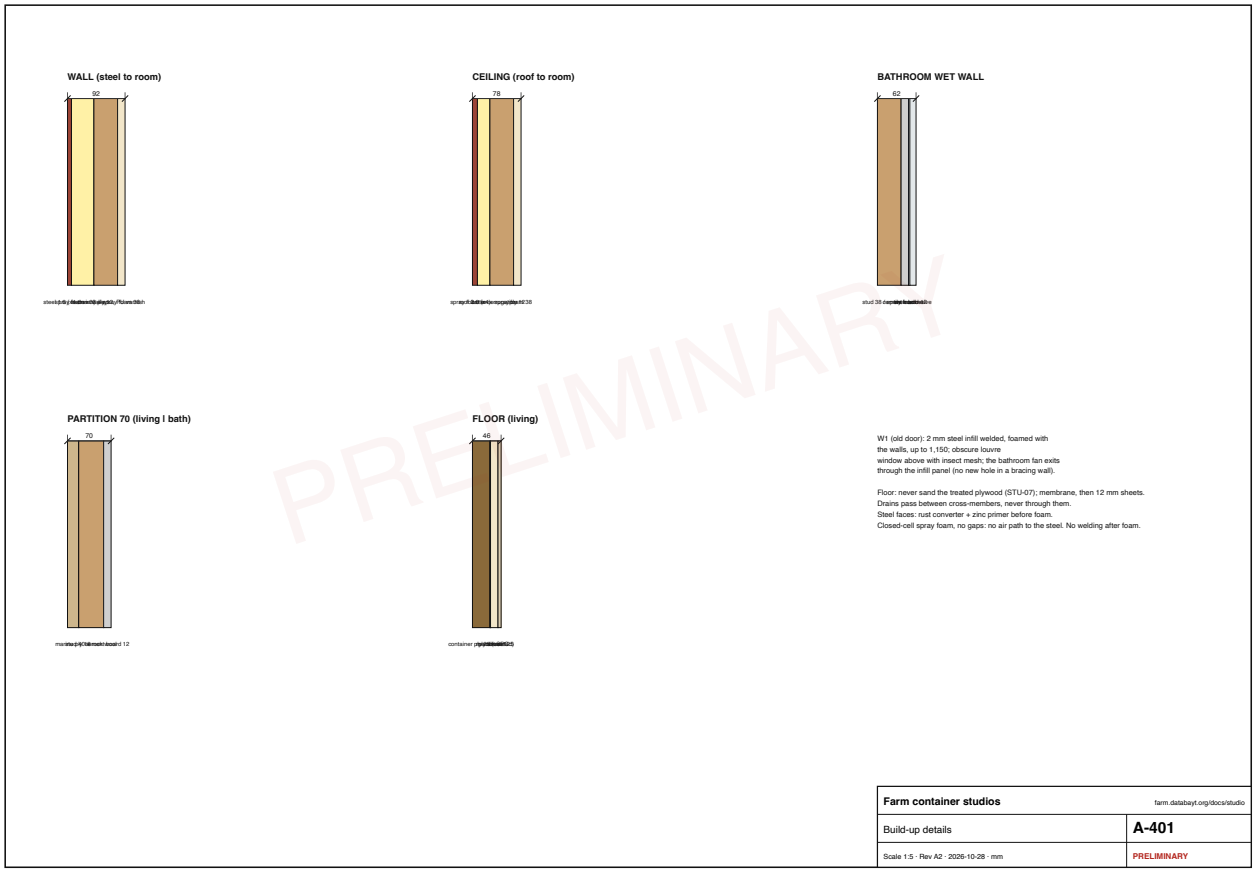
Ibishushanyo



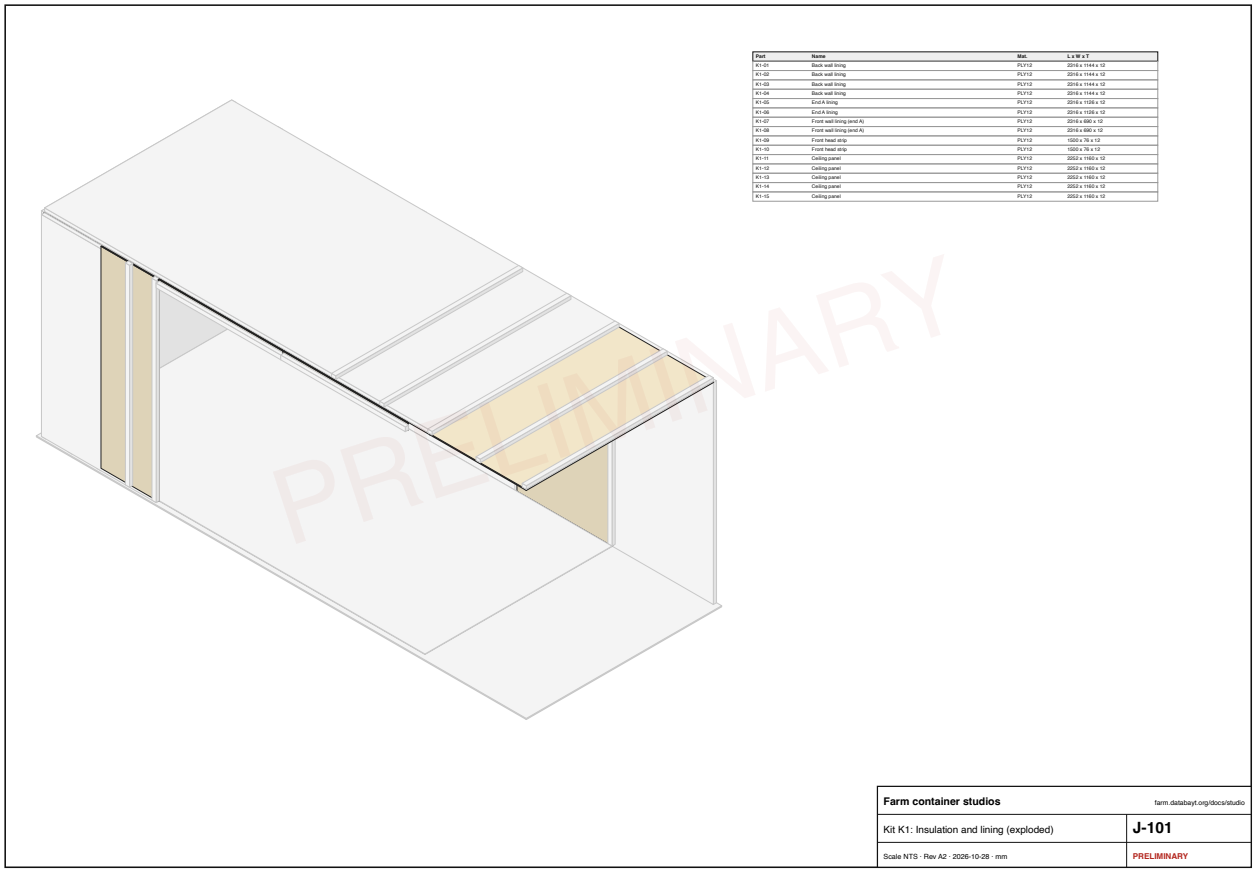
A-103 Furniture and dimension plan · open full size



A-201 Interior elevations: back wall and front wall · open full size



A-401 Build-up details · open full size



J-101 Kit K1: insulation and lining · open full size

Ibindi bishushanyo byose, AS-01 kugeza AS-27 na C-101 kugeza C-107, biri ku rutonde rw'ibishushanyo.